ECA5601 –Advanced Wireless Communication Systems / 5G / 6G
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Experiment: 7

Analog Beamforming Analysis Using MATLAB DATE:12-8-26

Aim:To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objectives:

To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.

Software Required: • MATLAB

Theory:

Analog beamforming is a beam steering technique in which a single RF chain drives multiple antenna elements through adjustable phase shifters. By controlling the phase of the transmitted or received signals, the antenna array forms a directional beam toward the intended user, thereby improving signal strength, coverage, and spectral efficiency. Analog beamforming is widely employed in millimeter-wave (mmWave) communication systems due to its reduced hardware complexity and lower power consumption.

MATLAB Code:

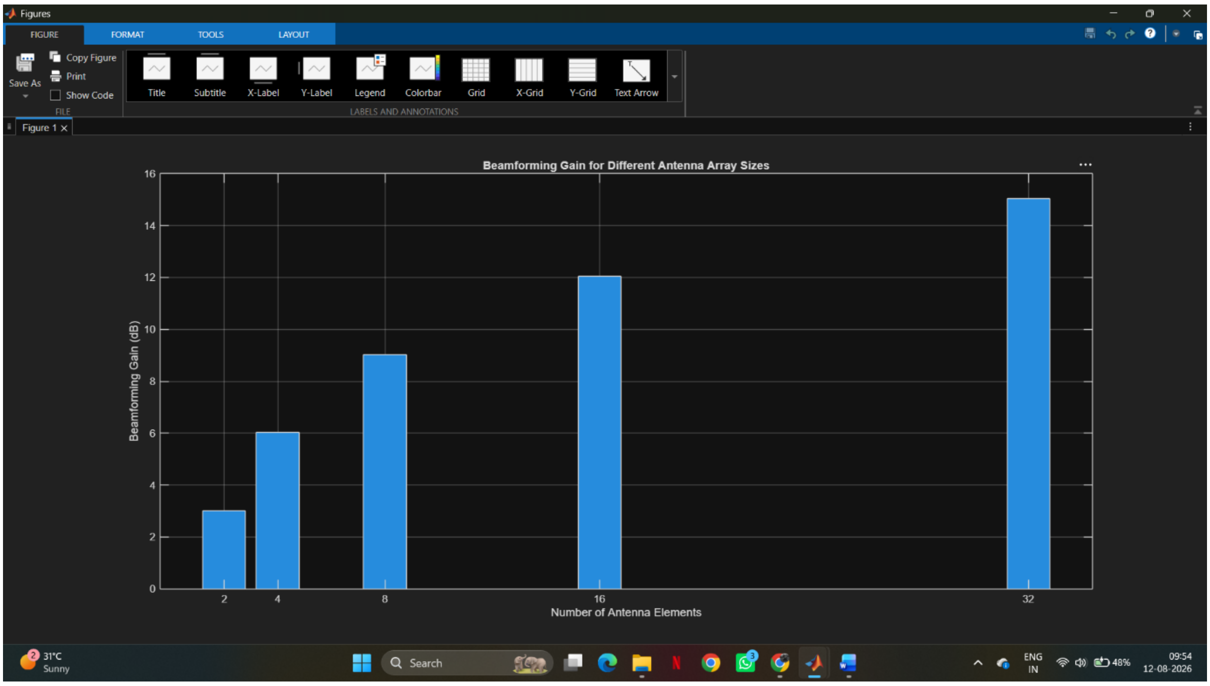
```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on
```

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

title('Beamforming Gain for Different Antenna Array Sizes')

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.